

WEIYI HUANG

(512) 902-6038
US Citizen, ITAR eligible

wh474@cornell.edu
[linkedin.com/in/weiyi-huang8](https://www.linkedin.com/in/weiyi-huang8)

EDUCATION

Cornell University, College of Engineering, Bachelor of Science **Expected December 2027**

Electrical and Computer Engineering, Aerospace Engineering Minor

Relevant Courses: Microelectronics, Circuits, Computer Architecture, Probability for Random Signals, Thermodynamics, Statics and Mechanics of Solids, Fluid Mechanics; Signals and Systems, System Dynamics, Aeronautics (FA26)

ENGINEERING EXPERIENCE

MAVRIK, Inc. (acquired by HAVOC AI), Lead Electrical Engineer **May 2025-Aug 2026**

- Owned EE hardware roadmap through **startup acquisition** by HAVOC AI (Feb 2026).
- Integrated and brought up avionics interfaces with CubePilot FC, including ESCs (CAN/PWM), u-blox GPS (I2C/UART), RC, telemetry, and companion compute; debugged interface issues through bench and flight testing.
- Architected a 6-layer Raspberry Pi CM5 carrier board deployed with 200+ flight-hours; implemented USB 3.0, power delivery, Ethernet EMI mitigation, and controlled-impedance routing within NDAA and SWaP-C constraints.
- Designed a custom 4-in-1, ~2 kW peak ESC; revised layout, switching behavior, and MOSFET/gate-resistor selection after bench testing to reduce case temperature by ~20%; debugged low-level STM32 C firmware over SWD.
- Mentored two summer EE interns with design reviews on three PCB designs now deployed in the field.

Selected Vehicle Programs:

Hatchet (~80 kW peak electric propulsion, Group 3 UAS)

- Owned electrical architecture of Group 3 UAS that won Army xTech Overwatch and SBIR Phase I contract (Oct 2025).
- Troubleshoot and rebuilt CAN bus architecture using Saleae and Kvaser Leaf due to low noise margin and telemetry dropouts during flight. Increased message reception rate with new topology by ~60% and eliminated overload events.
- Traced an ESC HV thermal runaway with shipment in five days to QC and internal capacitor defects supplied by the manufacturer. Reconstructed harness with added capacitance within 72 hours and cleared vehicle to fly on schedule.
- Laid out the wire harness topology for the FC, power, and payload systems. Revised harness for improved reliability and safety by introducing LV bulkhead connectors and an HV busbar (before previously mentioned thermal runaway).

Locust (vertically integrated, Group 1 UAS)

- Consolidated CV compute and flight-critical LV/HV systems onto two PCBs for system-level integration (DFM). Finalized connector placement enabled plug-and-play assembly and cut drone manufacturing time by ~60%.
- Made systems-level tradeoffs across PCBs in Ethernet speed (1000BASE-T to 100BASE-TX), USB wire harnessing (USB port to locking connector), and power requirements (isolated peripheral and compute power regulation) to size each interface to mission requirements than available headroom. Drove final design of customer-facing hardware.

Cornell University AutoDrone, Electrical Subteam Lead

Sept 2024-Present

- Built the electrical systems for high-speed "LawnDart" UAS, including RX/TX, FPV, and the high-power architecture (50V, 10 kW peak); debugged avionics, HV power, and INAV firmware to achieve reliable flight tests and consistent benchmarks for 250+ mph world record attempts.
- Engineered a custom motor test stand using an Arduino-based embedded system and supporting sensors. Characterized thrust, RPM, and EMI noise with automated testing for steady-state and stepped throttle conditions.

LEADERSHIP EXPERIENCE

Project Level the Field (Project LTF), Chapter Founder and President

May 2024-May 2025

- Founded Project LTF's Cornell chapter, a 501(c)(3) nonprofit providing college app resources to underserved students. Led 20-person team to develop a 12-week schedule and organized 2 workshops with local high schools.

Boy Scouts of America, Eagle Scout

Dec 2021

SPECIALIZED SKILLS

PCB & Hardware Design: KiCad/Altium EDA, controlled impedance, differential routing; high-speed interfaces (USB 3.0, Ethernet); power delivery/DC-DC, fault-tolerant power, EMI mitigation, ESC/high power design, hardware testing/validation, firmware configuration. **Embedded & Avionics:** C/C++, Python, Verilog, Bash; STM32/Cortex-M7 and ATtiny microcontrollers, Raspberry Pi CM; CAN, UART, SPI, I2C, PWM; bus utilization analysis, Cube/ArduPilot, INAV, Mission Planner, GPS/external IMU, telemetry. **Bring-Up & Testing:** Oscilloscopes, spectrum analyzers, logic analyzers (Saleae), CAN (Kvaser), DMM, MCU debuggers; EMI/RF testing, motor test stand; LTspice, SolidWorks, ANSYS; Linux, Git/GitHub, JIRA.

**All values above are approximates or approved to prevent any NDA violations.*